Opposing Modulation of EEG Aperiodic Component by Ketamine and Thiopental: Implications for the Noninvasive Assessment of Cortical E/I Balance in Humans

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Abstract

The balance between excitatory and inhibitory (E/I) activity is critical for brain function, and its disruption is implicated in neuropsychiatric disorders. Electrophysiological signals can be decomposed into periodic (oscillatory) and aperiodic components. In the power spectrum, the periodic component appears as narrowband peaks, while the aperiodic component underlies its characteristic $1/f^{\chi}$ power-law decay. Computational models predict that shifts in E/I balance alter the exponent χ in specific directions. In a randomized, double-blind, placebo-controlled, within-subject study, healthy volunteers received subanesthetic doses of ketamine and thiopental during an EEG oddball task. These drugs have opposite effects on E/I balance but comparable sedative profiles. Ketamine reduced the PSD exponent, while thiopental increased it, consistent with computational predictions. Changes in the exponent were associated with subjective and cognitive effects. These findings suggest that the PSD exponent has potential as a noninvasive EEG biomarker sensitive to transient shifts in cortical E/I balance.

Introduction

From the cellular to the global scale, brain circuits maintain a balance between excitatory and inhibitory processes[1]. Both aberrant increases and decreases in the ratio of excitation to inhibition (E/I) are associated with reduced circuit function[2] following an inverted-U relationship[3-6]. Alterations in E/I balance also characterize a wide range of brain disorders, including epilepsy, autism, and schizophrenia[7-11], where they are associated with impairments in cognitive function. Thus, understanding the biology of disorder- and drug-induced alterations in E/I balance could shed light on the neurobiology of psychiatric disorders and their treatment.

In the frequency domain, the power spectral density (PSD) of global electrophysiological brain signals such as local field potentials (LFPs), electroencephalography (EEG), and magnetoencephalography (MEG), decays with frequency (f) following a power law of the form $1/_{f^{\chi}}$ $(\chi > 0)$[9, 12, 13]. Evidence from in silico, animal, and human studies suggests that this power law arises from an aperiodic, non-oscillatory, broadband component of neural field signals that emerges from the sum of the globally uncoordinated and mainly stochastic activity of large neural populations[14]. In addition, brain signals have a periodic component which corresponds to the well-known neural oscillations that emerge from recurrent cycles of coordinated activation and deactivation of large populations of neurons[14]. This component appears as narrowband peaks in the power spectrum[12, 13], and is critically involved in brain functions such as perception, attention, and working memory[15]. Alterations in this component are associated with mental illness, neurological conditions, and drug intoxication[16, 17]. In contrast, the role of the aperiodic component remains poorly understood. Although it emerges from stochastic processes, it is modulated by task demands[18, 19], and thus cannot be easily equated with noise in the traditional sense of task-unrelated activity.

Pharmacological studies in humans with drugs that transiently shift the balance between excitation and inhibition have demonstrated that shifts in E/I balance have significant effects on the periodic component (e.g., through event-related potentials and the auditory steady-state response)[20-23], and that these effects correlate with the cognitive and perceptual effects of the drugs. In contrast, the effects that changes in E/I balance have on the aperiodic component and their relationship with behavior have received little attention in human studies. Studies have been conducted on humans under general anesthesia, which precludes the study of the subjective effects associated with controlled shifts in the PSD exponent and, more importantly, are affected by the confounding of loss of consciousness, which has profound effects on the exponent χ of the PSD curve[24-26] . In these studies, propofol, a drug that increased γ-aminobutyric acid type A receptor (GABAA-R) mediated circuit inhibition, has been shown to increase the PSD exponent, while ketamine, an N-methyl-D-aspartate glutamate receptor (NMDA-R) antagonist with disinhibitory effects at low doses, has been shown to decrease it[27, 28]. Recently, the reanalysis of data from two small MEG pharmacological studies in non-anesthetized humans showed that reducing circuit inhibition decreased the PSD exponent compared to placebo (n=19), and that increasing inhibition increased the PSD exponent compared to placebo (n=15)[26]. No direct, between-study comparison of active drug conditions was conducted, nor was the relationship between changes in E/I balance and subjective effects of the drugs reported.

These findings are consistent with evidence from computer modeling and animal studies showing that changes in E/I balance are inversely correlated with the PSD exponent χ, particularly between 30-50Hz [29]; i.e., the higher the E/I ratio, the smaller the χ (the flatter the PSD curve). However, simulations have shown that factors other than E/I ratio also have a significant influence on the PSD exponent, and that under certain conditions, the proposed relationship may not hold[14]. For example, a study in rodents failed to reduce the PSD exponent after chemogenetically reducing $GABA_A$-R activity[27, 30]. Thus, to isolate the effects that changes in E/I balance have on the aperiodic component, it is essential to maintain other factors influencing the PSD exponent as constant as possible across compared conditions. These challenges and those associated with studying anesthetized subjects, could be significantly addressed with a within-subject design administering subanesthetic doses of drugs with opposite effects on E/I balance but similar behavioral profiles. To the best of our knowledge, this is the first study to implement this approach.

In this study, we addressed the confounding effects of interindividual variability and alertness using a randomized, double-blind, placebo-controlled, within-subject design, using an EEG oddball task to examine the effects of acute, drug-induced changes in E/I ratio on the aperiodic component of brain signals and their relationship with subjective effects and cognitive function. For this purpose, we administered subanesthetic doses of ketamine and thiopental, sedative drugs with opposite effects on E/I ratio but comparable behavioral profiles, to a large sample of healthy participants. Ketamine is a dissociative anesthetic[31] and a non-competitive antagonist of the NMDA-R. Ketamine blocks NMDA-R-mediated excitation[32], and reduces glutamate release[33], particularly at anesthetic doses. However, at subanesthetic doses, ketamine produces a paradoxical cortical disinhibition by preferentially reducing NMDA-R-mediated excitatory input to inhibitory interneurons[34-36], leading to increased glutamate release[33] and increased high frequency pyramidal cell firing, albeit in a chaotic pattern associated with a suppression of bursting activity [37, 38]. The effects of ketamine on cortical microcircuits contrast with those of another sedating medication, thiopental; a barbiturate positive allosteric modulator of $GABA_A$-Rs[20, 39]. Thiopental reduces cortical excitation[40] and decreases glutamate release[41]. These drugs were selected for having opposite effects on E/I ratio at subanesthetic doses and comparable sedative effects[20, 42] thus, alterations in EEG measures would not be due to asymmetric changes in arousal[43]. We hypothesized that the increase in E/I-ratio induced by ketamine would reduce the PSD exponent, while the reduction in E/I ratio induced by thiopental would increase it. Furthermore, we hypothesized that the magnitude of the changes in E/I ratio would be correlated with the subjective effects and cognitive impairment of the drugs as measured by the Clinician-Administered Dissociative States Scale (CADSS)[44] and Hopkins Verbal Learning Test (HVLT).

To better understand the relevance of the aperiodic component in capturing E/I ratio changes in the brain, we conducted data-driven analyses to assess the relative importance of the PSD exponent for identifying altered E/I states (drug conditions) against a large set of commonly used EEG measurements. We hypothesized that the PSD exponent would be the most informative biomarker for distinguishing drugs with opposite effects on E/I ratio, suggesting that changes in the aperiodic component may serve as a sensitive index of aberrant shifts in E/I balance, especially in within subject designs.

Results

Behavioral measures

The analyses revealed a significant main effect of drug condition on the CADSS total score (ATS=222.664, p<0.001). Post hoc comparisons indicated significantly higher scores in the ketamine condition compared to both placebo and thiopental (all p_{Adj} <0.001), and higher scores in the thiopental condition compared to placebo (p_{Adj}<0.001). For the HVLT average score, there was a significant main effect of drug condition (ATS=44.599, p<0.001). Post hoc comparisons showed significantly lower scores under ketamine compared to both placebo and thiopental (all p_{Adj} <0.001), and lower scores under thiopental compared to placebo (p_{Adj}=0.001) (see Table 1 and Figure 1).

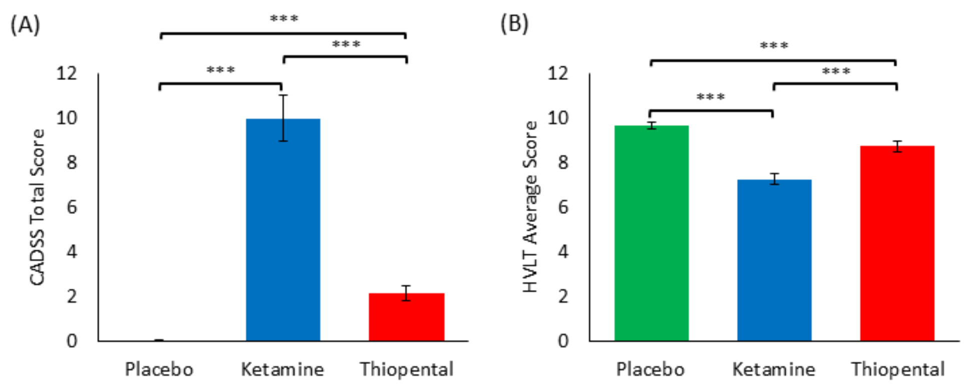

Figure 1. Behavioral effects across drug conditions.
(A) CADSS total scores across placebo, ketamine, and thiopental conditions. (B) HVLT average scores across the same conditions. Horizontal lines and asterisks indicate significant pairwise differences (***p<0.001, Holm-Bonferroni corrected). Error bars represent the standard error of the mean (SEM).

Table 1 Behavioral effects across drug conditions

Drug Condition	CADSS Total	HVLT Average
Placebo	0.041 (0.200)	9.699 (1.206)
Ketamine	10.013 (8.813)	7.280 (2.021)
Thiopental	2.150 (2.543)	8.756 (1.902)

Means and standard deviations (SD) are reported for subjective effects (CADSS Total) and cognitive performance (HVLT Average) under each drug condition. Standard deviations are shown in parentheses.

EEG measures

Power spectral density

For illustrative purposes, the average EEG power spectral density (PSD) across subjects and electrodes is shown for the post-infusion placebo, ketamine, and thiopental conditions, as well as for the pre-infusion period (Figure 2).

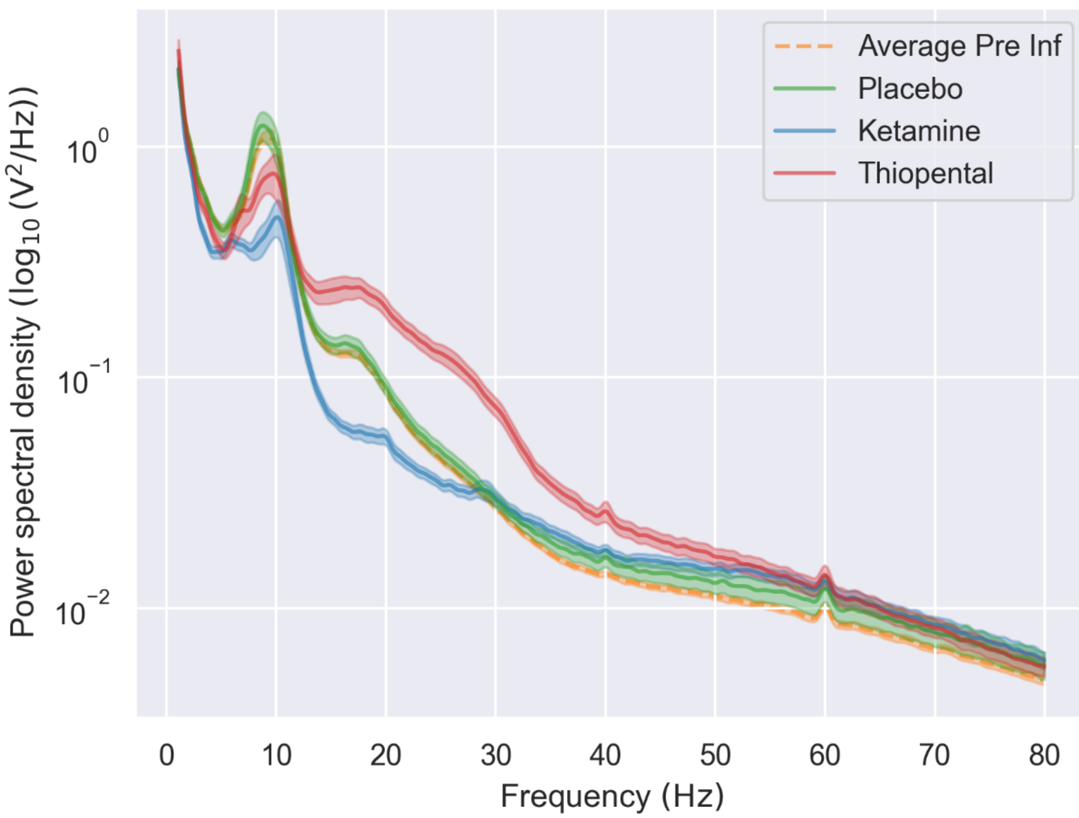

Figure 2. EEG power spectral density curves across drug conditions.
Power spectral density (PSD) curves ($\log_{10}$-transformed) averaged across subjects and electrodes are shown for the post-infusion placebo (green), ketamine (blue), thiopental (red), and pre-infusion placebo (dashed orange) conditions. PSDs were computed from the standard stimulus trials during the oddball task. Error shading represents the standard error of the mean.

Hypothesis-driven approach: Differences in PSD exponent across drug conditions

There was a main effect of drug on the absolute and the relative (change from pre-infusion to post-infusion) values of the PSD exponent during the pre-stimulation (baseline) and stimulation periods of the EEG task (p_{Adj}<0.001, adjusted for 4 comparisons). Pairwise comparisons revealed that ketamine reduced, and thiopental increased the PSD exponent compared to placebo in all cases (p_{Adj}<0.001, adjusted for 12 comparisons). For details, see Tables 2 and 3.

Table 2 Drug Effects on the PSD exponent across time windows

PSD exponent	Time window	Omnibus drug effect (Wald X^2, *df*=2)	*p*-value (Corrected)	Drug comparison	*p*-value (corrected)
Absolute (post-infusion)	Pre-stimulation	306.045	<0.0001	Ketamine<Placebo	<0.0001
				Ketamine<Thiopental	<0.0001
				Thiopental>Placebo	<0.0001
	Stimulation	389.323	<0.0001	Ketamine<Placebo	<0.0001
				Ketamine<Thiopental	<0.0001
				Thiopental>Placebo	<0.0001
Relative (change from Pre-infusion)	Pre-stimulation	216.366	<0.0001	Ketamine<Placebo	<0.0001
				Ketamine<Thiopental	<0.0001
				Thiopental>Placebo	<0.0001
	Stimulation	325.774	<0.0001	Ketamine<Placebo	<0.0001
				Ketamine<Thiopental	<0.0001
				Thiopental>Placebo	<0.0001

Results of the omnibus tests and pairwise comparisons of the effects of drug on absolute and relative (change from pre-infusion) PSD exponent values. For the omnibus tests, p-values were adjusted across 4 comparisons; for the pairwise comparisons, p-values were adjusted across 12 comparisons using the Holm-Bonferroni approach.

Table 3 Absolute and relative PSD exponent values across drug conditions and time windows

Drug condition	Mean PSD exponent (95% CI)			
	Absolute (post-infusion)		Relative (change from pre-infusion)	
	Pre-stimulation	Stimulation	Pre-stimulation	Stimulation
Placebo	2.170 (2.025, 2.315)	2.174 (2.013, 2.335)	-0.096 (-0.187, -0.005)	-0.076 (-0.165, 0.013)
Ketamine	1.483 (1.342, 1.625)	1.386 (1.240, 1.533)	-0.758 (-0.864, -0.653)	-0.873 (-0.973, -0.772)
Thiopental	2.839 (2.673, 3.005)	2.871 (2.699, 3.044)	0.607 (0.461, 0.753)	0.686 (0.548, 0.824)

Marginal means and 95% Wald confidence intervals (CI) are reported for the PSD exponent, estimated using generalized estimating equations (GEE). Results are shown separately for absolute (post-infusion) and relative (change from pre-infusion) values across pre-stimulation and stimulation time windows.

For illustrative purposes, we show the relative location of the subjects across drug conditions in a 3-dimensional space, with the PSD exponent on the X-axis and the change from pre-infusion in the PSD exponent on the Y-axis (Figure 3).

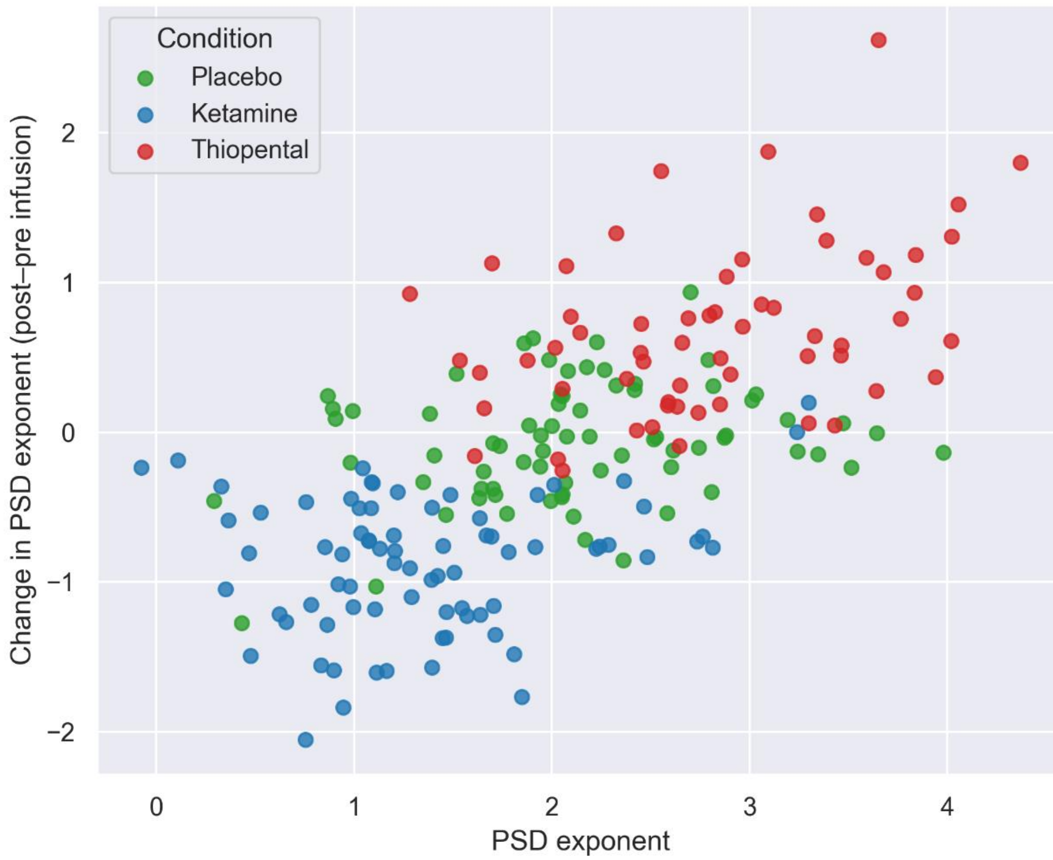

Figure 3. PSD exponent values per subject across drug conditions.
Each point represents an individual subject. The X-axis shows the PSD exponent during post-infusion, while the Y-axis shows the change in PSD exponent from pre- to post-infusion. PSD exponents were calculated during the stimulation interval of the standard trials of the oddball task. Colors indicate drug condition (placebo, ketamine, thiopental).

Relationship between PSD exponent and behavior

The analyses showed significant positive associations between the PSD exponent and both the CADSS total (log_{10}-transformed; Figure 4A) and HVLT average scores (Figure 4B), for both absolute (post-infusion) and relative (change from pre-infusion) PSD exponent estimates, across both pre-stimulation and stimulation periods of the oddball task (Table 4).

Table 4 Relationship between PSD exponent values and behavioral outcomes

Behavioral outcome	PSD exponent	Time window	Beta	R^2_{marg}	*p*-value (corrected)
CADSS total (log scale)					
	Absolute (post-infusion)	Pre-stimulation	–0.536	0.260	<0.0001
		stimulation	-0.550	0.262	<0.0001
	Relative	Pre-stimulation	-0.483	0.226	<0.0001

	(change from pre-infusion)				
		Stimulation	-0.505	0.248	<0.0001
HVLT average					
	Absolute (post-infusion)	Pre-stimulation	0.301	0.094	0.0001
		Stimulation	0.309	0.093	0.0001
	Relative (change from pre-infusion)	Pre-stimulation	0.268	0.083	0.0003
		Stimulation	0.273	0.082	0.0004

Beta coefficients, marginal R^2 values, and Holm-Bonferroni-corrected p-values (8 comparisons) from the GEE models assessing the relationship between PSD exponent values and behavioral outcomes across the ketamine and thiopental conditions. PSD exponents were computed for the post-infusion period (absolute) and as the difference between post- and pre-infusion PSD exponents (relative). These measures were calculated separately for the pre-stimulation and stimulation periods of the oddball task. Behavioral outcomes include the CADSS total ($\log_{10}$-transformed) and HVLT average scores.

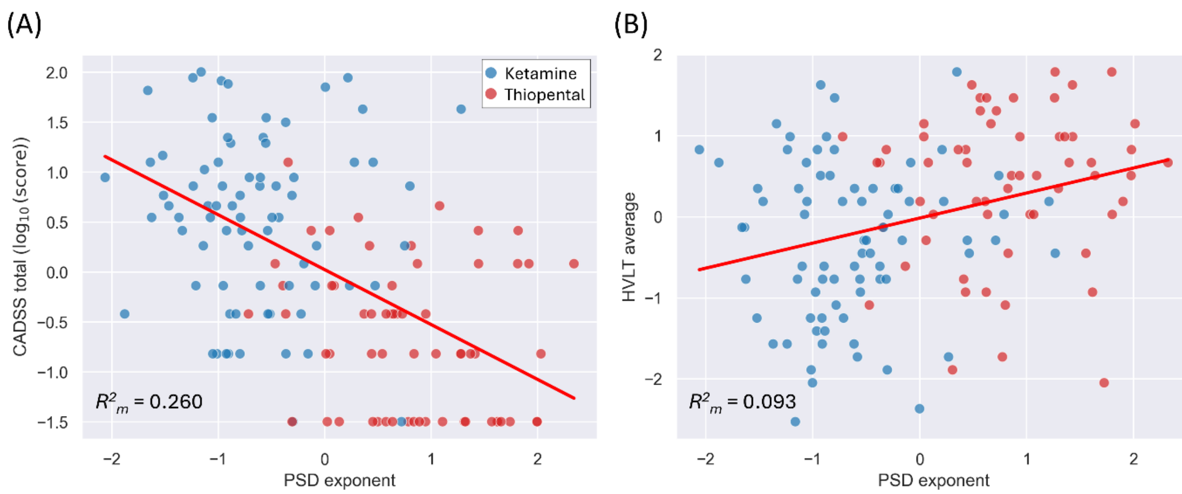

Figure 4. Relationship between PSD exponent and behavioral outcomes across active drug conditions.
(A) Each dot represents an individual subject's standardized (z-score) CADSS total score ($\log_{10}$-transformed) as a function of the standardized PSD exponent during the stimulation period of standard trials. (B) Same as in (A), but with HVLT average scores. The red line indicates the GEE-based longitudinal linear regression fit across pooled data from the ketamine and thiopental conditions. Marginal R^2 values (R^2_m) are shown in the bottom-left corner of each plot.

Data-driven approach: Identification of informative EEG biomarkers for differentiating drug conditions

The EEG biomarker categories identified as the most informative for distinguishing among drug conditions are presented in Table 5. The full lists of selected biomarker categories, including those that did not meet the 50% cross-validation selection frequency threshold, are presented in Supplementary Tables S6-S9, grouped by drug comparison and selection size restriction parameter alpha. The list of biomarkers within each category is provided in Supplementary Table S1. The full lists of the selected individual EEG biomarkers (i.e., not grouped under categories) are presented in Supplementary Tables S2-S5. Classification performance metrics for each drug condition comparison and level of alpha are provided in Supplementary Tables S10.1-S10.4.

Table 5 Most informative EEG biomarker categories for drug condition discrimination

Drug Comparison	Number of Selections (Folds)	PSD Exponent (Aperiodic)	Delta Power 1st Peak (Periodic)	Theta Power (Total)	Beta Power (Total)
Ketamine vs Thiopental vs Placebo	20	**100%** (20)	**50%** (10)	**60%** (12)	**65%** (13)
Ketamine vs Placebo	20	**95%** (19)	**60%** (12)	5% (1)	**65%** (13)
Thiopental vs Placebo	20	**65%** (13)	30% (6)	**90%** (18)	**75%** (15)
Ketamine vs Thiopental	20	**100%** (20)	0% (0)	5% (1)	**50%** (10)
All Comparisons	80	**90%** (72)	35% (28)	40% (32)	**63.75%** (51)

The table presents the EEG biomarker categories that were selected in at least half (≥10/20) of the cross-validation folds for at least one of the drug condition comparisons listed in the first column. For each comparison, the values to the right indicate the percentage of folds in which each category was selected, with the number of selections in parentheses (20 folds total: 5 per alpha level). Bolded numbers identify the categories selected in at least 50% of the folds for each comparison. The final row shows the number of selections (80) across all the comparisons.

Relationship between selected EEG biomarkers and behavior

The feature selection algorithm identified four EEG biomarker categories, comprising 24 individual biomarkers. Of these, 13 biomarkers belonging to the PSD exponent, Beta power (total), and Peak delta Power (periodic) categories were significantly (corrected) associated with the CADSS total score (log_{10}-transformed). The strongest associations were observed for members of the PSD exponent category (largest β=-0.550, p_{Adj}<0.0001), followed by members of the beta power (total) category (largest β=-0.454, p_{Adj}<0.0001), and a member of the peak delta power (periodic) category (largest β=-0.308, p_{Adj}=0.0022). For the HVLT total score, four biomarkers belonging to the PSD exponent category showed significant (corrected) associations (largest β=-0.309, p_{Adj}< 0.0202). For the full list of the regression results, see Supplementary Table S11.

Discussion

In this randomized, placebo-controlled study, we examined the effects that transient, pharmacologically induced shifts in cortical E/I balance have on the aperiodic component of EEG signals, and how these effects relate to subjective drug experiences and cognitive performance. We assessed the aperiodic component by estimating the exponent of the power spectral density (PSD) from EEG signals collected during an oddball task. To manipulate E/I balance, we intravenously administered subanesthetic doses of ketamine and thiopental, agents with comparable sedative profiles known to transiently increase and decrease E/I ratio, respectively. In addition, we used a data-driven, information theory-based approach to evaluate how information relevant for differentiating between drug conditions was carried by the PSD exponent compared to a broad range of commonly used EEG and MEG measures.

We found that under ketamine, the PSD exponent was reduced (i.e., flatter PSD curve) compared to both placebo and thiopental, and that under thiopental, it was increased relative to placebo (i.e., steeper PSD curve). These effects were consistent across absolute and relative (change from pre-infusion) values, and across the pre-stimulation and stimulation periods of the oddball task. As expected, both ketamine and thiopental increased perceptual alterations compared to placebo, with ketamine increasing them more than thiopental. The regression analyses revealed that in the active drug conditions, lower PSD exponent values were associated with greater subjective effects. Similarly, ketamine and thiopental induced verbal memory deficits compared to placebo, with ketamine inducing larger deficits than thiopental. Regression analyses also showed that lower PSD exponent values were associated with greater disruptions in verbal memory performance.

Our data-driven analyses identified four EEG biomarker categories that were consistently selected in at least one drug condition comparison. These results suggest that among all the biomarker categories, these ones carried the most relevant information for distinguishing between drug conditions under selection size constraints. Among these, the PSD exponent was the most consistently selected category across comparisons, suggesting that it carried a comparatively greater amount of useful information for distinguishing drug conditions. Total beta power ranked second in overall consistency and therefore in the amount of useful information it carried. In contrast, in the pairwise comparisons, peak delta power was informative for differentiating between ketamine and placebo only, while total theta power was informative for differentiating between thiopental and placebo only.

Our results in awake humans showed that a transient, drug-induced increase in E/I balance reduced the PSD exponent, whereas a transient, drug-induced reduction in E/I balance increased it. These findings provide the first evidence in awake humans, using drugs with comparable sedative effects, to support recent modeling work proposing an inverse relationship between E/I balance in cortical circuits and the PSD exponent of global electrophysiological signals[29]. While these results suggest that the PSD exponent may serve as a noninvasive measure sensitive to shifts in E/I balance in humans, especially in within-subject designs, they must be replicated using other compounds with known effects on E/I balance to assess their generalizability and limitations.

A wide range of EEG biomarkers commonly used to characterize drug-induced changes in brain dynamics failed to reach the consistency (informativeness) threshold we set in our feature selection algorithm. As a result, well-established measures such as signal-to-noise ratio and Lempel-Ziv complexity were excluded by our approach. Importantly, these results do not imply that biomarker categories that were not selected by our algorithm captured no relevant or unique drug-related information, but that under pressure to differentiate between the drug conditions with as few biomarkers as possible, the selected ones stood out as carrying the largest amount of relevant information.

Variability in the selected biomarkers across pairwise comparisons suggests that some variables carried information about drug-induced changes in E/I balance that was not fully captured by the changes in the PSD exponent. Specifically, total beta power was selected as informative in every pairwise comparison of drug conditions. While the interpretation of these findings goes beyond the scope of this article, which focuses on the interaction between E/I balance and the aperiodic component, it is important to highlight that total beta power was considered informative across comparisons. This coincides with the well-known effects of thiopental increasing[45] and ketamine suppressing[46, 47] power in the beta band. In addition, periodic delta power was useful for distinguishing ketamine from placebo, and total theta power was useful for distinguishing thiopental from placebo. The value of these measures as informative biomarkers to differentiate between states of increased and decreased E/I balance is something that will need to be explored in future studies.

Ketamine, an NMDA receptor antagonist, produces dissociative anesthesia at higher doses (2mg/kg) and rapid antidepressant effects at lower doses (0.5-1.0mg/kg)[48, 49]. In this study, a subanesthetic reduced the PSD exponent, supporting the hypothesis that ketamine's disinhibitory effects are associated with a shift in EEG spectral power from lower to higher frequencies[50]. Preclinical studies have shown that ketamine increases E/I balance in the medial prefrontal cortex (mPFC) by reducing interneuron activity, as demonstrated using electrophysiology[51-53], microdialysis[54], and other approaches[55]. However, intracortical recordings in nonhuman primates have shown that ketamine's effects vary across neuronal populations and cortical layers[56]. While ketamine disinhibits pyramidal cells in cortical layer 5 (L5), it inhibits them in cortical layer 3 (L3)[56]. Interestingly, recent modeling work using biophysically realistic neurons showed that L5 pyramidal cells are by far the largest contributor to EEG signals[57]. Therefore, our results showing a reduction in the PSD exponent, which is consistent increased E/I balance, is likely reflecting the ketamine's disinhibitory effects on L5, as its inhibitory effects on L3 are likely masked[56].

Thiopental acts by enhancing GABA-mediated inhibition of synaptic transmission, leading to a decrease in E/I balance. In this study, we observed an increase in the PSD exponent following thiopental administration, consistent with preclinical and human studies showing that GABAergic agents increase the PSD exponent[27, 30].

At subanesthetic doses, ketamine and related NMDA receptor antagonists increase the E/I ratio by disinhibiting cortical circuits, while thiopental and other GABAergic agents reduce the E/I ratio by enhancing inhibition[20, 39, 51-53]. Despite having opposite effects on E/I balance, they both induce significant impairment in brain function[58-61 62], suggesting an inverted-U relationship between E/I balance and brain circuit output. Deviations in either direction from an optimal task-related E/I balance would lead to reduced neural efficiency[3-6]. The nature of the impairment depends on the direction of the shift in E/I balance. Computational models suggest that elevated E/I balance can lead to impulsive responses, while reduced E/I balance leads to indecisive responses in decision-making tasks[6]. Consistently, in this study, we observed that both ketamine-induced reductions (increased E/I ratio) and thiopental-induced increases (reduced E/I ratio) in the PSD exponent were associated with increased subjective effects and verbal memory impairment. This is consistent with previous studies showing that the PSD exponent tracks neurocognitive performance across domains[27, 63, 64], including verbal learning tasks[65]. However, our results also showed that ketamine's subjective and cognitive effects were stronger than those of thiopental, indicating an asymmetry in the inverted-U relationship. Thus, at comparable sedation levels, the substances showed that circuit disinhibition induced more severe subjective and cognitive disruptions than hyperinhibition.

The subjective effects of ketamine have been linked to its antidepressant action, such that higher CADSS scores at 40 minutes post-infusion correlate with reductions in depressive symptoms at both 230 minutes and 7 days post-infusion. Considering the positive correlation between CADSS scores and PSD exponent observed in this study, the association between CADSS scores and antidepressant effects raises the possibility that the PSD exponent may help identify patients who are more likely to benefit from ketamine treatment. If future studies confirm this relationship, it could eventually help reduce the economic, social, and personal burden associated with current trial-and-error treatment strategies.

There are several limitations to this study. It is important to note that E/I balance can be conceptualized at various scales, from excitatory and inhibitory inputs onto a single neuron to long-range neural networks[66]. The global scale of EEG signals cannot disentangle the specific changes occurring at the cellular or neuroreceptor level. Other approaches, including magnetic resonance spectroscopy (MRS)[67] and neuroimaging[10, 68], as well as transcranial magnetic stimulation (TMS) paradigms, such as short-interval intracortical inhibition (SICI), intracortical facilitation (ICF), and long-interval intracortical inhibition (LICI)[69], offer alternative markers of E/I balance (e.g., receptor occupancy, neurotransmitter metabolism, and concentration) in humans. For example, pharmacological fMRI has been used to generate empirical models of E/I balance based on functional connectivity[70, 71]. Furthermore, cerebrospinal fluid GABA levels have been used as a proxy measure for E/I balance alterations in patients with schizophrenia[72-74]. However, these methods are relatively intrusive, expensive, and/or require specialized infrastructure. Additionally, their poor temporal resolution makes it difficult to capture the fast dynamics of E/I balance. Nevertheless, parallel studies using these approaches in humans in combination with pharmacological agents with known effects on E/I balance may help to better understand and identify the electrophysiological fingerprints of illness- or drug-induced alterations in E/I balance. Currently, there is a range of promising electrophysiological measures such as microstates, entropy, and beta and gamma power[1, 75, 76].

While the changes in the PSD exponent induced by ketamine and thiopental were consistent with their effects on E/I balance[51-53], we cannot conclusively rule out the contribution of factors unrelated to excitation and inhibition, such as drug-induced changes in the biophysical properties of neurons[77]. Furthermore, ketamine affects multiple neurotransmitter systems, including the opioid, aminergic, and cholinergic systems[78], making it difficult to determine how each pathway contributes to changes in the electrophysiological signals.

A further limitation concerns our data-driven biomarker selection approach, which may disregard mechanistically relevant measures. The method is designed to maximize mutual information with the drug condition labels while minimizing the size of the selected biomarker set. As a result, when multiple biomarkers carry overlapping information, the algorithm tends to retain only the most informative one, excluding others that may be less informative but still relevant for understanding the neural mechanisms underlying the effects of ketamine and thiopental.

Conflicts of interest

D.C.D. serves on the Scientific Advisory Board of Ceruvia Lifesciences and the Physicians Advisory Board of the Connecticut Medical Marijuana Program. He has served as a Consultant for Atai, Abide Therapeutics, Jazz Pharmaceuticals, and Biohaven. He is listed as an inventor on patent US20210236523A1 and serves on the Scientific Advisory Board of the Shulgin Institute. J.H.K has stock or stock option (with or without additional compensation) in Biohaven Pharmaceuticals, Cartego Therapeutics, Damona Pharmaceuticals, EpiVario Inc, Neumora Therapeutics, Rest Therapeutics, Response Pharmaceuticals, Tempero Bio, Terran Biosciences, and Tetricus Inc outside the submitted work. In addition, he reported serving as a paid scientific advisor for AbbVie, Aptinyx Inc, Biogen, Bionomics, Boehringer Ingelheim Pharmaceuticals, Cerevel Pharmaceuticals, Epiodyne, Eisei Pharmaceuticals, Jazz Pharmaceuticals, Johnson & Johnson, Novartis Pharmaceuticals, Psychogenics Inc, and Takeda Pharmaceuticals. Also, he has the following patents: US Patent No. 8,778,979 B2, US Patent No. 9592207, and U.S. Provisional Patent Application No. 62/719,935.

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Figure Legends

Figure 1. Behavioral effects across drug conditions.
(A) CADSS total scores across placebo, ketamine, and thiopental conditions. (B) HVLT average scores across the same conditions. Horizontal lines and asterisks indicate significant pairwise differences (***p<0.001, Holm-Bonferroni corrected). Error bars represent the standard error of the mean.

Figure 2. EEG power spectral density curves across drug conditions.
Power spectral density (PSD) curves ($\log_{10}$-transformed) averaged across subjects and electrodes are shown for the post-infusion placebo (green), ketamine (blue), thiopental (red), and pre-infusion placebo (dashed orange) conditions. PSDs were computed from the standard stimulus trials during the oddball task. Error shading represents the standard error of the mean.

Figure 3. PSD exponent values per subject across drug conditions.
Each point represents an individual subject. The X-axis shows the PSD exponent during post-infusion, while the Y-axis shows the change in PSD exponent from pre- to post-infusion. PSD exponents were calculated during the stimulation interval of standard trials from the oddball task. Colors indicate drug condition (placebo, ketamine, thiopental).

Figure 4. Relationship between PSD exponent and behavioral outcomes across active drug conditions.
(A) Each dot represents an individual subject's standardized (z-score) CADSS total score ($\log_{10}$-transformed) as a function of the standardized PSD exponent during the stimulation period of standard trials. (B) Same as in (A), but with HVLT average scores. The red lines indicate the GEE-based longitudinal linear regression fits across pooled data from the ketamine and thiopental conditions. Marginal R^2 values (R^2_m) are shown in the bottom-left corner of each plot.